BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6467

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. A flower is a modified:

(A) Shoot
(B) Leaf
(C) Root
(D) Petal

Q2. Example of free living fresh water flat worm is:

(A) Dugesia
(B) Fasciola
(C) Taenia
(D) Hydra

Q3. Pseudocoelom body cavity is found in:

(A) Ascaris
(B) Earthworm
(C) Neries
(D) Mosquito

Q4. Photosystem I has chlorophyll 'a', which absorbs maximum light of:

(A) 400nm
(B) 500nm
(C) 600nm
(D) 700nm

Q5. Glycolysis is the breakdown of glucose upto the formation of:

(A) Lactic acid
(B) Alcohol
(C) Pyruvic acid
(D) Acetic acid

Q6. Which has parasitic nutrition?

(A) Cuscuta
(B) Mycorrhiza
(C) Nitrogen fixing bacteria
(D) Lichens

Q7. When an oxygen tension is 115mm mercury, how much haemoglobin is saturated in percentage?

(A) 92%
(B) 94%
(C) 96%
(D) 98%

Q8. Pressure flow theory was proposed by:

(A) Ernst Munch
(B) Sacks
(C) Dixon
(D) Hook

Q9. Leucaemia is the result of uncontrolled production of:

(A) Leucocytes
(B) Thrombocytes
(C) Erythrocytes
(D) Platelets

Q10. In deductive reasoning we move from:

(A) General to general
(B) General to specific
(C) Specific to specific
(D) Specific to general

Q11. Which is not carbohydrate?

(A) Wood
(B) Cotton
(C) Paper
(D) Wax

Q12. An enzyme with its co-enzyme or prosthetic group removed is designated as:

(A) Holo enzyme
(B) Co-enzyme
(C) Apoenzyme
(D) Activator

Q13. Which is not found in secondary wall?

(A) Salts
(B) Silica
(C) Chitin
(D) Cellulose

Q14. Which is an insect?

(A) Cray fish
(B) Silver fish
(C) Jelly fish
(D) Star fish

Q15. Bacteria without any flagella is called:

(A) Atrichous
(B) Lophotrichous
(C) Monotrichous
(D) Peritrichous

Q16. Entamoeba histolytica causes in humans:

(A) Sleeping sickness
(B) Amoebic dysentery
(C) Malaria
(D) Cholera

Q17. It is non-hyphal unicellular fungi:

(A) Bacteria
(B) Rust
(C) Yeast
(D) Smut

Biology (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6467

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Define peptide and polypeptide bond.

(ii) Define apoenzyme and holoenzyme.

(iii) Differentiate between binding and catalytic site of enzyme.

(iv) What do you mean by induce fit model? Who proposed it?

(v) Write the ecological importance of fungi.

(vi) Define spore and conidia.

(vii) Write about Hookworm and pinworm.

(viii) Differentiate between acoelomate and coelomate.

(ix) Write about some affinities which echinoderms show with hemichordata.

(x) Differentiate between ostia and osculum.

(xi) What is the source of oxygen during photosynthesis?

(xii) Define carotenoids. How they are helpful in photosynthesis?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Differentiate between physiology and morphology.

(ii) Differentiate between freshwater Biology and marine Biology briefly.

(iii) What are Golgi apparatus? Give its function.

(iv) What are cristae and polysome?

(v) Write two differences between fungi and oomycotes.

(vi) What are kelps? Give their importance.

(vii) Write the importance of algae.

(viii) Give two examples of unicellular green algae.

(ix) Differentiate between microphyll and megaphyll.

(x) What is alternation of generations? How it is important for plant life?

(xi) Differentiate between diffusion and osmosis.

(xii) What are lymph nodes? What is their function?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Differentiate between mumps and measles.

(ii) Define Pilli. Also give its function.

(iii) What is trypsin?

(iv) What is saprophytic nutrition?

(v) Differentiate between villi and microvilli.

(vi) Define lung capacities.

(vii) What is myoglobin?

(viii) Define tuberculosis.

(ix) How pH affects haemoglobin to combine with oxygen?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Write a note on biological method.

Q5(b). (Marks 04) Discuss any eight functions of blood.

Q6(a). (Marks 04) Discuss in detail primary structure of proteins.

Q6(b). (Marks 04) Write a note on animal diseases caused by fungi.

Q7(a). (Marks 04) Discuss growth and reproduction in bacteria.

Q7(b). (Marks 04) Describe evolution of seed in plant.

Q8(a). (Marks 04) Describe the life cycle of bacteriophage.

Q8(b). (Marks 04) Make a sketch of non-cyclic phosphorylation.

Q9(a). (Marks 04) What are lysosomes? Give their importance with special emphasis on Tay-Sach's disease.

Q9(b). (Marks 04) Give the role of stomach in digestion of food.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6468

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The body cavity of Nematoda is called as:

(A) Blastocoel
(B) Haemocoel
(C) Pseudocoel
(D) Coelom

Q2. The phylum which is exclusively marine is:

(A) Echinodermata
(B) Pisces
(C) Porifera
(D) Amphibia

Q3. In respiratory chain, the coenzyme Q is oxidized by cytochrome:

(A) Cyt-b
(B) Cyt-a
(C) Cyt-ab
(D) Cyt-c

Q4. The end product of glucose breakdown in glycolysis is:

(A) Acetate
(B) Pyruvate
(C) Oxalate
(D) Fumarate

Q5. The inactive pepsinogen is secreted by cells:

(A) Oxyntic
(B) Zymogenic
(C) Mucous
(D) Gastric

Q6. The lungs are covered by double-layered thin membranous sacs called:

(A) Pleura
(B) Alveoli
(C) Air sacs
(D) Pericardium

Q7. Types of blood cells which stay from 10 - 20 hours in the blood are called:

(A) Monocytes
(B) Lymphocytes
(C) Neutrophils
(D) Leukocytes

Q8. The substance which inhibit blood clotting is:

(A) Histamine
(B) Heparin
(C) Disprin
(D) Fibrin

Q9. Pasteurization is widely used for preservation of:

(A) Vegetable
(B) Fruit
(C) Meat products
(D) Milk products

Q10. The branch of biology which deals with the study of chemical component and chemical process in living organism is called:

(A) Biochemistry
(B) Biogenetic
(C) Bioecology
(D) Biology

Q11. The competitive inhibitor of succinic acid is:

(A) Succinic acid
(B) Fumaric acid
(C) Citric acid
(D) Malonic acid

Q12. Which is not found in primary wall?

(A) Cellulose
(B) Hemicellulose
(C) Lignin
(D) Pectic

Q13. A disease, which is highly contagious is:

(A) Measles
(B) Mumps
(C) Herpes
(D) AIDS

Q14. Grape-like cluster of cocci-bacteria is called as:

(A) Diplococcus
(B) Streptococcus
(C) Staphylococcus
(D) Diplobacillus

Q15. Which is not a ciliate?

(A) Paramecium
(B) Vorticella
(C) Stentor
(D) Trypanosoma

Q16. Which is used to give flavour, aroma and colour to cheese?

(A) Yeast
(B) Puccinia
(C) Penicillium
(D) Agaricus

Q17. All seed producing plants are called:

(A) Bryophytes
(B) Tracheophytes
(C) Pteridophytes
(D) Spermatophytes

Biology (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6468

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Write briefly the protective function of water.

(ii) Differentiate between coenzyme and prosthetic group.

(iii) Define inhibitors. Give an example.

(iv) What is induce fit model of enzyme action?

(v) Differentiate between spore and conidia.

(vi) What are predator fungi? Give an example.

(vii) Differentiate between radial and bilateral symmetry.

(viii) What are parabronchi? Write their function.

(ix) What are nematocysts? Give their function.

(x) What is placenta and its role?

(xi) Why Calvin cycle is also called as C_3 pathway.

(xii) Define chemiosmosis.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Define embryology.

(ii) What is biotechnology?

(iii) What is unit membrane model of cell membrane?

(iv) What is primary cell wall? Give its chemical composition.

(v) Define kingdom Protista.

(vi) How zooflagellates obtain their food?

(vii) Give two examples of brown algae.

(viii) What is chlorella? Where it is found?

(ix) What is alternation of generation?

(x) What is double fertilization?

(xi) What are plasmodesmata?

(xii) Define transpiration.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Give biological classification of corn.

(ii) Write four phases in bacterial growth curve.

(iii) Distinguish between saprophytic and parasitic nutrition.

(iv) Name various types of salivary glands and give ingredients of saliva.

(v) Differentiate between Diarrhoea and Constipation.

(vi) What is Asthma? Give its effects.

(vii) Differentiate between haemoglobin and myoglobin.

(viii) Give composition of inhaled and exhaled air.

(ix) Give two factors which affect capacity of haemoglobin to combine with oxygen.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Write note on protection and conservation of environment.

Q5(b). (Marks 04) Write any eight functions of blood.

Q6(a). (Marks 04) Write a note on carbohydrates.

Q6(b). (Marks 04) Discuss two mutualistic symbiotic associations of fungi.

Q7(a). (Marks 04) Describe physical and chemical methods to control micro-organisms.

Q7(b). (Marks 04) Describe different adaptations of bryophytes to land habitat.

Q8(a). (Marks 04) Write a note on life cycle of AIDS.

Q8(b). (Marks 04) Write non-cyclic phosphorylation.

Q9(a). (Marks 04) Write a note on plastids.

Q9(b). (Marks 04) Discuss nutrition in insectivorous plants.